

Generation Interconnection System Impact Study Report

For Queue Project Q17-0757

Madera 69 kV
November 22, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0757. The Interconnection Customer proposes a 163.0 MW wind facility interconnecting at the Madera 69 kV substation in the North Carolina area. The request was received on March 10, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 269 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$10,714,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ROANOKE RAPIDS / MCCARTNEY 345.0 kV line	GASTON / ROANOKE RAPIDS 345.0 kV	114.3	1,398	9.9
HAGGIN / WALTHALL 138.0 kV	HAGGIN / N TOWER TAP 345.0 kV line	111.7	439	7.2
SUNNBROOK 138 kV/1 kV Transformer Bank	MCCABE / LAUREL CITY 138.0 kV line	109.9	181	9.9
MAGNOLIA / COSMAR 138.0 kV L	MAGNOLIA / CALIENTE 138.0 kV	109.7	217	7.1
PHELON PARTS / SUNNYBROOK 138.0 kV	MCCABE / LAUREL CITY 138.0 kV Line	103.8	188	9.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ROANOKE RAPIDS / MCCARTNEY 345.0 kV	Rebuild line with higher-capacity conductor	\$85,011,000	1% (\$860,000)
HAGGIN / WALTHALL SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$4,905,000	4% (\$175,000)
SUNNBROOK 138 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$5,587,000	100% (\$5,587,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MAGNOLIA / COSMAR 138.0 kV L	Reconductor line	\$2,609,000	11% (\$298,000)
PHELON PARTS / SUNNYBROOK 138.0 kV	Reconductor line	\$3,794,000	100% (\$3,794,000)
	Total allocated to project		\$10,714,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.